

# Unique decomposition Theorem.

For any  $B \in GL_n(\mathbb{C})$ , there exists a unique lower-triangular  $M$  with positive diagonal s.t  $MB$  is unitary.

Proof. Since  $B$  is invertible, the row vectors of  $B$ ,  $\{\beta_1, \dots, \beta_n\}$  is basis for  $V$ .

Apply Gram-Schmidt process, that is, for each  $1 \leq k \leq n$ , put

$$\alpha_k = \beta_k - \sum_{j < k} \frac{\langle \beta_k, \alpha_j \rangle}{\|\alpha_j\|^2} \alpha_j$$

Then,  $\text{span}\{\alpha_1, \dots, \alpha_r\} = \text{span}\{\beta_1, \dots, \beta_r\}$  and  $\{\alpha_1, \dots, \alpha_n\}$  ( $r \leq n$ ) is orthogonal.

Put  $U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \dots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix}$ , then clearly  $U^* U = I$ , because  $\{\alpha_1, \dots, \alpha_n\}$  is orthogonal.

And, for each  $1 \leq k \leq n$ ,  $\alpha_k = \beta_k - \sum_{j < k} C_{kj} \beta_j$  for some  $C_{kj}$ .

Put  $M_{kj} = \begin{cases} -\frac{C_{kj}}{\|\alpha_k\|} & \text{if } k > j \\ \frac{1}{\|\alpha_k\|} & \text{if } k = j \\ 0 & \text{otherwise.} \end{cases}$  Then, each  $1 \leq k \leq n$ ,  $U_k = \begin{pmatrix} & & 0 \\ & \ddots & \\ M_{k1} & \dots & M_{kn} \end{pmatrix} \cdot (\beta_1 \dots \beta_n) = U$

so  $MB = U$ . To show the uniqueness, denote

$$T^+(n) = \{M \in M_{nn}(\mathbb{C}) \mid M \text{ is lower triangle, positive diagonals}\}.$$

Let  $M_1, M_2 \in T^+(n)$  satisfies  $M_1 B, M_2 B$  are unitary. Since unitary matrices forms a group

$$(M_1 B) \cdot (M_2 B)^{-1} = M_1 M_2^{-1} \text{ is also unitary.}$$

And,  $M_2^{-1}$  is lower-triangular with positive diagonals, so is  $M_1 M_2^{-1}$ . Now,

$$(M_1 M_2^{-1})^{-1} = (M_1 M_2^{-1})^*, \text{ so } M_1 M_2^{-1} \text{ is diagonal.}$$

So,  $M_1 M_2^{-1} = I$ , or  $M_1 = M_2$ . @

Corollary. For each  $B \in GL(n)$ , there exists unique matrices  $K, A, N$  s.t  $B = KAN$ , where  $K$  is unitary,  $A$  is diagonal with positive values,  $N$  is upper-triangular with diagonals 1

+Note. Let  $T_n(F)$  be the set of  $n \times n$  upper-triangular matrices with all diagonals are 1.

Then,  $(T_n(F), \cdot)$  is group, because: for any  $A \in T_n(F)$ ,  $A = I + N$ ,

where  $N$  is strictly upper triangular with  $N^n = 0$ , nilpotent. So,

$$A^{-1} = I - N + N^2 - N^3 + \dots + (-N)^{n-1} \left( = \sum_{k=0}^{n-1} (-N)^k \right)$$

and  $A^{-1} \in T_n(F)$  is clear.

# QR - Factorization.

Let  $A = (w_1 \dots w_n)$  be an invertible matrix.

And, for each  $1 \leq k \leq n$ , 
$$v_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} \cdot v_i.$$

Then,  $\{v_1, \dots, v_n\}$  is an orthogonal, so put  $u_k = v_k / \|v_k\|$ , then

$\{u_1, \dots, u_n\}$  is an orthonormal. Clearly,  $Q \stackrel{\text{def}}{=} (u_1 \dots u_n)$  is unitary.

Moreover, for each  $1 \leq k \leq n$ ,

$$w_k = \|v_k\| \cdot u_k + \sum_{i=1}^{k-1} \langle w_k, u_i \rangle \cdot u_i. \left( = v_k + \sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} \cdot v_i \right)$$

Define

$$R_{jk} = \begin{cases} \|v_k\| & j=k \\ \langle w_k, u_j \rangle & j < k \\ 0 & j > k \end{cases}$$

Then,  $R$  is upper-triangular, and satisfying

$$Q^{-1}A = Q^*A = \begin{pmatrix} \overline{u_1} \\ \vdots \\ \overline{u_n} \end{pmatrix} \cdot (w_1 \dots w_n) = \begin{pmatrix} \langle w_1, u_1 \rangle & \langle w_2, u_1 \rangle & \dots & \langle w_n, u_1 \rangle \\ \langle w_1, u_2 \rangle & & & \vdots \\ \vdots & & & \\ \langle w_1, u_n \rangle & - & - & \langle w_n, u_n \rangle \end{pmatrix} \\ = R.$$

so,  $A = QR = QDR'$

$$\text{where } D = \begin{pmatrix} \|v_1\| & & & 0 \\ & \|v_2\| & & \\ & & \ddots & \\ 0 & & & \|v_n\| \end{pmatrix} \text{ and } R' = \begin{pmatrix} 1 & & & \\ & 1 & \langle w_k, u_j \rangle_{j < k} & \\ & & \frac{1}{\|v_3\|} & \\ 0 & & & 1 \end{pmatrix}$$